SUPPORTING ADULT GUIDANCE

Changing the subject

Formulae and Rearranging · Lesson 5. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson teaches changing the subject of a formula in which the new subject appears once. Learners run function machines backwards, rearrange on a level balance that refuses out-of-order steps and shows a numerical check on every line, find faulty steps in worked rearrangements, and turn rearranged formulae into converters for many values.

Driving Question: *How do I turn a formula round to find what I need?*

The mathematics, in plain terms

Changing the subject applies inverse operations to both sides of the formula in the reverse order to that in which they act on the intended subject. Every intermediate line is equivalent to the original, which can be confirmed numerically by substituting values that satisfy the original. Where the subject is squared and represents a length, the positive square root is taken.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Two steps

Make t the subject of v = u + at.

1. What happens to t? First × a, then + u.	$t \rightarrow \times a \rightarrow + u \rightarrow v$
2. Undo + u first: subtract u from both sides.	$v - u = at$
3. Then undo × a: divide both sides by a.	$(v - u)/a = t$
4. Check with u = 2, a = 3, t = 4: v = 14.	$(14 - 2)/3 = 4$ ✓

t = (v – u)/a.

Example 2 — Rearrange, then use

F = 1.8C + 32 converts °C to °F. Make C the subject and convert 212 °F.

1. Subtract 32 from both sides.	F – 32 = 1.8C
2. Divide both sides by 1.8.	C = (F – 32)/1.8
3. Substitute F = 212.	C = 180/1.8 = 100

C = (F – 32)/1.8, so 212 °F is 100 °C.

Why the balance refuses some steps

The balance only applies the step that undoes the last operation. If a learner chooses to divide first in v = u + at, it explains that dividing now would divide u as well. That step is not wrong in itself, but it leads to a harder route. This keeps the reverse-order idea clear at Foundation level. Year 10 meets cases where the subject appears twice.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Reverse the machine: build the backwards machine with inverse steps in reverse order.	~7 min
Movement	The formula balance: undo steps on both sides, with a numerical check on every line.	~15 min
Reflection	Spot the faulty rearrangement, with an optional numerical test.	~9 min
Creativity	Build a converter: rearrange, then use the formula for many values at once.	~9 min
Rest	Reverse order in everyday life: the last thing on is the first thing off.	~2 min
Nutrition	Intellectual: flexibility — reshaping a tool to answer a new question.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Undoing in the wrong order (dividing before subtracting in $y = 3x + 5$).
Draw the machine: which step happened last?

Dividing only part of a side ($y/3 = x + 5$).
Test with numbers: $x = 2$, $y = 11$. Is 11/3 = 7?

Using the operation instead of its inverse (v + u = at).
Ask: 'What undoes + u?'

Multiplying instead of dividing to undo a coefficient ((F − 32) × 1.8).
Check with F = 77, which should give C = 25.

Leaving the subject on both sides, or the wrong letter alone.
Ask the learner to read the final line aloud: 'C equals …' — is C on its own?

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: solving one- and two-step equations, and inverse operations. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes $y/3 = x + 5$ as the first step for $y = 3x + 5$. The aim is a question that lets them find the slip themselves.

- "What happens to x first, and what happens last?"
- "Try $x = 2$: what is y? Is your line still true?"

- “When you divide the right-hand side by 3, what happens to the 5?”
- “Which step should you undo first?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Are other correct forms accepted?
Yes. All checking is by value, so $x = (y + 3)/2$, $x = y/2 + 1.5$ and $(y + 3) \div 2 = x$ are all accepted.

Why does the balance refuse a valid step?
To keep the focus on reverse order. The explanation shows why that step leads to a harder route.

What about square roots?
The stretch makes s the subject of $A = s^2$. As s is a length, only the positive root is used.

What comes next?
Lesson 6 uses and checks formulae in context: deciding whether to substitute or rearrange, and whether an answer makes sense.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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